Political Determinants of Government Budget Allocation in Thailand

SASATRA SUDSAWASD

AMORNRAT APINUNMAHAKUL

School of Development Economics

National Institute of Development Administration (NIDA)

Motivation of the study

- Military forces play a crucial role in the political development of Thailand.
- 11 out of 29 Prime Ministers are appointed PMs.
- 14 out of 29 PMs have strong military background
- Most of the elected governments are coalition of various allied parties.

Research question

- Whether the political development of the country would have any affects on the patterns of Thailand public expenditures?

- We answer the question by running two hypotheses testing

  1) Military influence hypothesis.

  2) Coalition parties influence hypothesis.

Literature review

1) Political budget cycle theory:

In order to increase their probabilities of being re-elected, the incumbent governments are likely to stimulate the economy prior to an election (Nordhaus, 1975).

During the election years, public expenditures are shifted more towards current spending rather than the capital investment, as the former are easier to notice and appreciated by voters while the later take years before its outcomes be realized (Rogoff and Sibert 1988; Vergne 2009).

Literature review

2) Partisan theory:

As different political party may pursue different economic goals, the contradict policy objectives could lead to different patterns of public expenditures (Hibbs 1977; Alesina 1987).

As the decision costs increase with the number of decision makers, the size of government expenditures can be influenced by the number of governing parties in a coalition (Weingast et al 1981).

Literature review

3) Swing voters versus rewarding loyalty hypotheses:

Studies in the US found that aligned states received more fund from the federal government than unaligned states (Larcinese et al. 2006; Grossman 1994; Levitt and Synder 1995; Solé-Ollé and Sorribas-Navarro 2008).

On the opposite, Federal government also manipulated government funds to target marginal states with more volatility of the presidential vote in the elections (Wright 1974; Wallis 1987)

Related literature review

- This study focus on the changes of the composition of expenditure budgets allotted towards each administrative ministry during the fiscal period of 1980-2013 rather than the amount of public expenditure of each ministry over time.

- Assuming that each coalition party would negotiate for more expenditure budget share for the ministries that the party was in charged, we hence categorized public spending using the administrative classification rather than the functional classification.

Some Major Findings

- The results indicate an indirect effect of a military coup and weak political institutions on the science and technological advancement of the country especially on the agricultural development.

- Military background of the prime minister also plays a crucial role in Thai government budget formulation. The prime ministers with military background behave differently from their civilian counterparts.

- As the percent share of MP of the leader party increase, more public spending is geared towards the Central Fund for the PM and less to the Office of the Prime Minister.

Model specification

A variation of Vergne (2009)'s model on the allocation of public spending is employed to examine the relationship between a number of political characteristics and the expenditure share to each line ministry.

$$Budget_{i,t} = \beta_0 + \beta_1 Budget_{i,t-1} + \beta_2 X_t + \beta_3 Political_{i,t} + \varepsilon_t$$

,where $Budget_{i,t}$ stands for ministry i's share of the planned expenditure budget in the fiscal year t.

X is the set of control variables.

Political is the set of political variables.

- Control variables (*X*)

Variable	Definition
GDPPC	Level of development, measured by the logarithm of real income per capita.
URBAN	Degree of urbanization, measured by the logarithm of the percentage of urban population.
OPEN	Degree of trade openness, measured by the ratio of exports plus imports to Gross Domestic Product (GDP).
TOT	Terms of trade.
AGE14	Young-age population aged between 0 and 14 years old as a percentage of the total population.
AGE65	Old-age population aged 65 and above as a percentage of the total population.

Variable	Definition
POP	Population.
UNEMPLOY	Unemployment rate.
TREND	Time trend variable.
YEAR2003	Year 2003 dummy variable, taking on the value of zero for years before 2003 and one for years 2003 and onwards (to capture the structural change of Thai ministries since 2003).
RECESSION	Recession dummy variable equals to one for the years: 1997, 1998, and, 2009 when Thailand entered into a recession period, and zero otherwise.

- Political variables of interest (*Political*)

Variable	Definition
Non-Elected Government	= 1 if a government is a non-elected government administration, = 0 otherwise.
MILITARY PM	= 1 if a prime minister is a former military officer, = 0 otherwise.
Share of MP_i	The percent share of Members in Parliament that minister i holds.
$LEADER_i$	= 1 if a minister i is from a quota of a prime minister or a member of a leader party in a multi-party coalition government or a member of a ruling single-party government, = 0 otherwise.
$OUTSIDER_i$	= 1 if a minister i is appointed from outside the parliamentary system, = 0 otherwise.

Military Influence Hypothesis

- *Non-elected government*

- During the fiscal years of 1980-2013, three out of the 18 administrative cabinets were post-coup governments. We will test whether a non-elected administration allocated public spending differently from their elected counterparts.

- It is likely that a post-coup government may provide more federal fund for the military spending in order to gain trust and support from the ruling junta, but how this affects other ministries has not yet been investigated.

- *Military background of a prime minister*

- Prime ministers with high ranks in the armed force may have a different viewpoint of how to run the country as compared to their civilian counterpart.

- Military PMs may divert more funds towards military projects and foreign affairs as opposed to the civilian PMs who might emphasize on different economic activities.

- Their consequences to economic development in the long run could differ significantly.

Coalition Party Influence Hypothesis

- *Share of Members in the Parliament*

- Share of MP of a partner party should indicate the party's negotiation power for executive positions in the Cabinet as well as the budget share for the ministers they are in charged.

- We will test whether or not the *Share of* MP_i has a positive influence on the composition of government expenditure at the administrative level.

- *Political Party of a Minister*

- To investigate whether the governing party would allocate more budget to *'reward'* their own loyal members or more for the partner party so as to 'strengthen' their coalition.

- In another words, ***'political party of a minister'*** variable can by use to test whether 'Rewarding loyalty' or 'Swing Voters' hypothesis is more suitable in the Thai context.

- *Outside minister*

- *Outsider ministers* are defined as those ministers appointed from "outside" the parliamentary system (or non-parliamentarians).

- *An outsider ministers* might lack of political power to negotiate for the budget as compared with those "inside" the parliamentary system

Data and Empirical Issues

- The budgeting data are the planned expenditure budget for 13 ministries plus the Central Fund for the prime minister during the period of 1980-2013. They are obtained from the annual report of *Thailand's Budget in Brief.*

- The macroeconomic data were extracted from the World Bank's *World Development Indicators*, the IMF's *International Financial Statistics*, and the *Penn World Table* 8.

- Data on military and political party variables derive from various sources ranging from newspapers, government websites, and government published documents (such as the Secretariat of the Cabinet and the Secretariat of the House of Representatives of Thailand, etc.)

- Since the disturbance terms are likely to be correlated across equations, the Seemingly Unrelated Regression Equations (SURE) introduced by Zellner (1962) is an appropriated estimation for this study.

- The augmented Dickey–Fuller test is performed for the unit root test. If a series has a unit root, then the first differences are taken.

- The Durbin alternative statistical test for serial correlation (when the model contains a lagged dependent variable) is employed. The test results cannot reject the null hypothesis of no first order serial correlation for each of the estimated equations at the 1 percent level of significance.

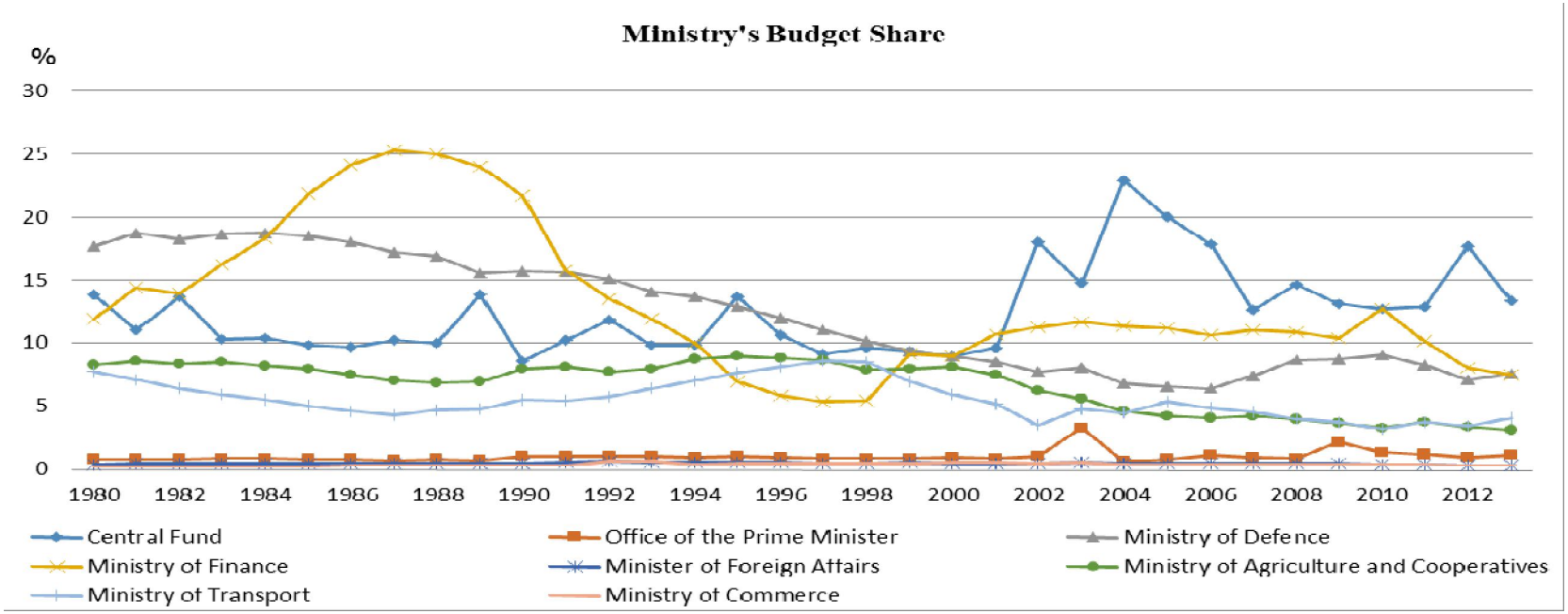
- The Breusch-Pagan test for heteroskedasticity is applied. The findings cannot reject the null hypothesis of homoscedasticity.

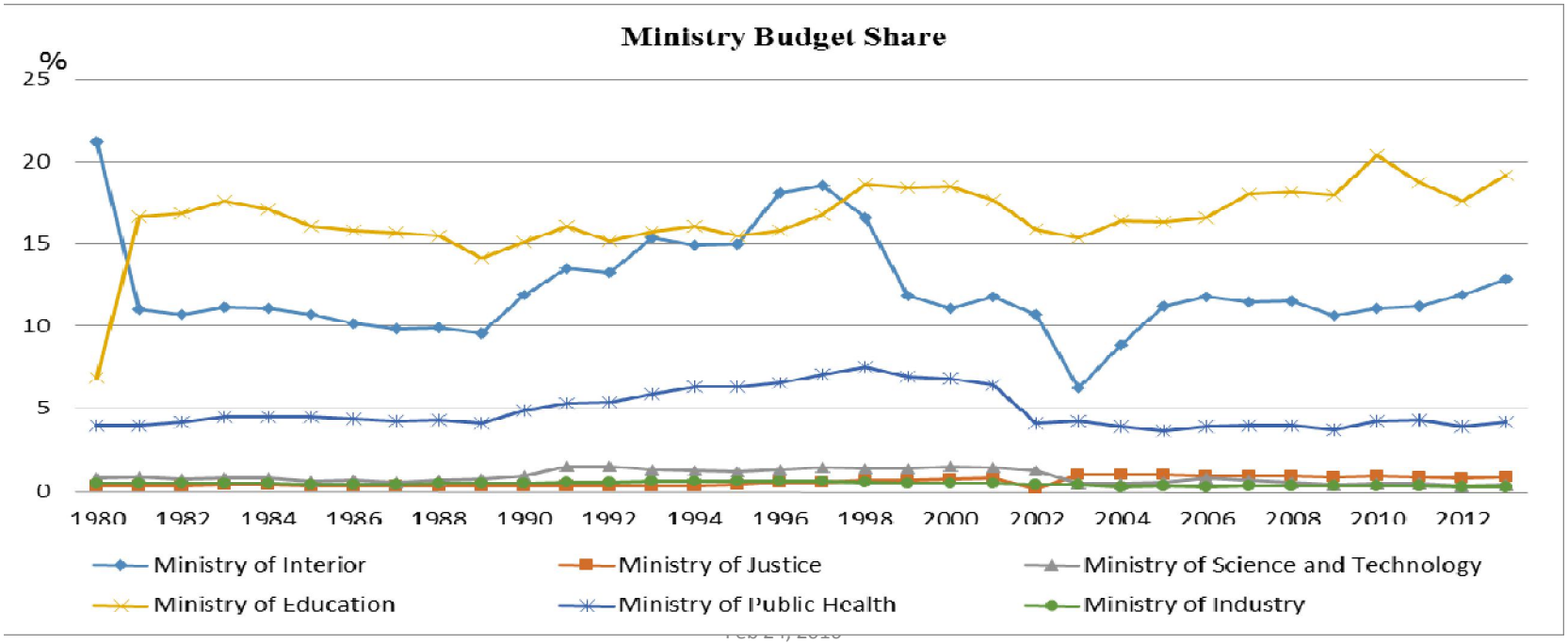
Some stylized facts

Ministry budget expenditures for the Fiscal Years (FYs) 1980 and 2013

Rank in FY 1980	Rank in FY 2013	Ministry	Ministry budget in FY 1980 (Million baht)	Ministry budget in FY 2013 (Million baht)	Share of budget in FY 1980 (in percentage of total budget	Share of budget in FY 2013 (in percentage of total budget)	Percentage change in share of budget (in FY 1980-2013)
1	3	Ministry of Interior	24,290	309,205	21.20	12.88	-8.32
2	4	Ministry of Defense	20,307	180,811	17.73	7.53	-10.19
3	2	Central Fund	15,848	319,207	13.83	13.30	-0.53
4	5	Ministry of Finance	13,639	179,249	11.91	7.47	-4.44
5	8	Ministry of Agriculture and Cooperatives	9,440	73,039	8.24	3.04	-5.20
6	7	Ministry of Transport	8,828	96,072	7.71	4.00	-3.70
7	1	Ministry of Education	7,863	460,075	6.86	19.17	12.31
8	6	Ministry of Public Health	4,495	100,153	3.92	4.17	0.25
9	9	Office of the Prime Minister	902	25,382	0.79	1.06	0.27
10	11	Ministry of Science and Technology	855	8,962	0.75	0.37	-0.37
11	14	Ministry of Industry	526	6,425	0.46	0.27	-0.19
12	12	Minister of Foreign Affairs	462	7,902	0.40	0.33	-0.07
13	10	Ministry of Justice	389	19,534	0.34	0.81	0.47
14	13	Ministry of Commerce	240	7,460	0.21	0.31	0.10
		Total planned budget	*114,556.5*	*2,400,000*			

Source: The data are obtained from the *Thailand's Budget in Brief* (various years).


Ministry's Budget Share
%
30
25
20
15
10
5
0
1980
1982
1984
1986
1988
1990
1992
1994
1996
1998
2000
2002
2004
2006
2008
2010
2012
Central Fund
Office of the Prime Minister
Ministry of Defence
Ministry of Finance
Minister of Foreign Affairs
Ministry of Agriculture and Cooperatives
Ministry of Transport
Ministry of Commerce


Ministry Budget Share
%
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1980
1982
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2002
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2006
2008
2010
2012
Ministry of Interior
Ministry of Justice
Ministry of Science and Technology
Ministry of Education
Ministry of Public Health
Ministry of Industry

- Descriptive statistics

	Obs.	Mean	S.D.	Min	Max
OUTSIDER$_i$					
Ministry of Defence	34	0.50	0.51	0	1
Ministry of Finance	34	0.59	0.50	0	1
Ministry of Foreign Affairs	34	0.32	0.47	0	1
Ministry of Agriculture and Cooperatives	34	0.15	0.36	0	1
Ministry of Transport	34	0.26	0.45	0	1
Ministry of Commerce	34	0.32	0.47	0	1
Ministry of Interior	34	0.44	0.50	0	1
Ministry of Justice	34	0.12	0.33	0	1
Ministry of Science and Technology	34	0.12	0.33	0	1
Ministry of Education	34	0.38	0.49	0	1
Ministry of Public Health	34	0.12	0.33	0	1
Ministry of Industry	34	0.32	0.47	0	1
LEADER$_i$					
Ministry of Defence	34	0.82	0.39	0	1
Ministry of Finance	34	1.00	0.00	1	1
Ministry of Foreign Affairs	34	0.68	0.47	0	1
Ministry of Agriculture and Cooperatives	34	0.41	0.50	0	1
Ministry of Transport	34	0.41	0.50	0	1
Ministry of Commerce	34	0.71	0.46	0	1
Ministry of Interior	34	0.85	0.36	0	1
Ministry of Justice	34	0.59	0.50	0	1
Ministry of Science and Technology	34	0.35	0.49	0	1
Ministry of Education	34	0.68	0.47	0	1
Ministry of Public Health	34	0.65	0.49	0	1
Ministry of Industry	34	0.41	0.50	0	1

	Obs.	Mean	S.D.	Min	Max
MP Share$_i$					
Central Fund	33	0.41	0.23	0.22	1.00
Office of the Prime Minister	33	0.41	0.23	0.22	1.00
Ministry of Defence	33	0.35	0.25	0.07	1.00
Ministry of Finance	33	0.41	0.23	0.22	1.00
Ministry of Foreign Affairs	33	0.36	0.26	0.05	1.00
Ministry of Agriculture and Cooperatives	33	0.24	0.28	0.04	1.00
Ministry of Transport	33	0.29	0.28	0.07	1.00
Ministry of Commerce	33	0.36	0.27	0.02	1.00
Ministry of Interior	33	0.38	0.25	0.07	1.00
Ministry of Justice	33	0.33	0.29	0.01	1.00
Ministry of Science and Technology	33	0.27	0.29	0.05	1.00
Ministry of Education	33	0.35	0.27	0.06	1.00
Ministry of Public Health	33	0.34	0.28	0.05	1.00
Ministry of Industry	33	0.26	0.30	0.01	1.00
Non-Elected Government	34	0.09	0.29	0	1
MILITARY PM	34	0.44	0.50	0	1

Empirical findings

Findings from the baseline specification:

- The basic determinants of budget share for each individual ministry are dissimilar.

- As Thailand becomes more developed, measured by an increase in the real income per capita, Ministry of Agriculture and Cooperatives receives a larger share of government budget.

- The degree of urbanization reduces the budget share allocated to the Ministry of Foreign Affairs, Ministry of Agriculture and Cooperatives, Ministry of Justice, and Ministry of Public Health respectively.

- An increase in total trade shares to GDP negatively affects the shares of government budget to Ministry of Foreign Affairs, while an increase in the terms of trade positively affects Ministry of Transport.

Findings (continued):

- As the young-age population ratio declines, budget composition of the Central Fund for the prime minister, the Office of the Prime Minister, and Ministry of Transport dwindle consequently. On the contrary, a rise in the old-age population ratio reduces the budget share of the Ministry of Justice.
- An increase in the number of population negatively affects the shares of public expenditure for the Ministry of Defense, Ministry of Agriculture and Cooperatives, Ministry of Interior, Ministry of Public Health, and Ministry of Industry.
- An increase in the unemployment rate has a positive implication on the government budget allocated to the Ministry of Education and the Ministry of Public Health. This finding supports the view that governments raise education budgets when unemployment surges.
- Ministry of Justice experienced a significant positive time-trend effect on its budget share.
- More budgetary fund is allocated to the Office of the Prime Minister during a recession period.

Baseline results

(Dependent variable: Ministry budget share in total government budget)

VARIABLES	(1) Central Fund (*Budget*)	(2) Office of the Prime Minister (*Budget*)	(3) Ministry of Defence (*D_Budget*)	(4) Ministry of Finance (*D_Budget*)	(5) Ministry of Foreign Affairs (*D_Budget*)	(6) Ministry of Agriculture and Co. (*D_Budget*)	(7) Ministry of Transport (*D_Budget*)
D_Budget$_{(t-1)}$ (or *Budget*$_{(t-1)}$)	0.0771	-0.199**	-0.271**	-0.0415	-0.350***	-0.144*	-0.214***
	(0.0653)	(0.0780)	(0.109)	(0.0668)	(0.0614)	(0.0792)	(0.0586)
*D*_log(*GDPPC*)	17.73	3.433	0.141	-20.77	-0.0046	5.940**	-1.807
	(19.30)	(3.119)	(4.292)	(13.34)	(0.302)	(2.928)	(4.078)
D_URBAN	2.395	-0.290	0.323	1.269	-0.0774*	-1.055***	-0.302
	(2.674)	(0.432)	(0.602)	(1.852)	(0.0419)	(0.402)	(0.567)
D_OPEN	0.0244	-0.0141	-0.0061	-0.0375	-0.0016*	0.0047	0.00766
	(0.0560)	(0.0091)	(0.0124)	(0.0388)	(0.0009)	(0.0084)	(0.0118)
TOT	-0.0519	-0.0119	-0.0533*	0.0355	-0.0018	-0.0288	0.0699***
	(0.125)	(0.0202)	(0.0281)	(0.0862)	(0.00196)	(0.0189)	(0.0264)
D_AGE14	9.727**	1.333**	-2.540***	-4.707*	0.0541	-1.102*	3.020***
	(3.916)	(0.638)	(0.887)	(2.723)	(0.0612)	(0.590)	(0.826)
AGE65	0.533	-0.297	0.198	-0.0293	-0.0375	0.493	-0.837
	(2.481)	(0.400)	(0.552)	(1.708)	(0.0387)	(0.377)	(0.523)
*D*_log(*POP*)	133.8	34.73	-122.8***	375.8***	2.191	-93.99***	-19.23
	(191.3)	(30.87)	(45.65)	(133.2)	(2.996)	(30.23)	(40.40)
D_UNEMPOY	-0.196	0.0016	0.0416	-0.210	0.0021	0.0443	-0.0259
	(0.424)	(0.0685)	(0.0943)	(0.293)	(0.0066)	(0.0643)	(0.0897)
TREND	-0.413	0.0276	-0.150	0.267	0.0038	-0.119	0.133
	(0.536)	(0.0865)	(0.122)	(0.370)	(0.0084)	(0.0821)	(0.113)
RECESSION	2.409	0.720*	-0.258	-2.481	-0.0485	0.0286	0.354
	(2.498)	(0.404)	(0.555)	(1.725)	(0.0391)	(0.375)	(0.527)
YEAR2003	4.738**	1.060***	0.433	-0.692	0.0945***	-0.0452	1.754***
	(2.412)	(0.373)	(0.510)	(1.587)	(0.0362)	(0.352)	(0.487)
Constant	22.08	3.999	6.234*	-15.11	0.396	2.793	-3.256
	(16.38)	(2.636)	(3.685)	(11.28)	(0.257)	(2.463)	(3.447)
Observations	32	32	32	32	32	32	32
R-squared	0.643	0.486	0.454	0.479	0.257	0.532	0.578

Note: Figures in parentheses are standard errors. *D_* denotes the first difference.
***, **, * indicate significance levels at 1%, 5%, and 10%.

VARIABLES	(8) Ministry of Commerce (*D_Budget*)	(9) Ministry of Interior (*D_Budget*)	(10) Ministry of Justice (*Budget*)	(11) Ministry of Science and Technology (*D_Budget*)	(12) Ministry of Education (*D_Budget*)	(13) Ministry of Public Health (*D_Budget*)	(14) Ministry of Industry (*D_Budget*)
$D_Budget_{(t-1)}$ (or $Budget_{(t-1)}$)	-0.307***	-0.0102	0.0949	0.0808	-0.00195	-0.277***	-0.323**
	(0.0998)	(0.0467)	(0.0684)	(0.0678)	(0.0413)	(0.0550)	(0.129)
*D*_log(*GDPPC*)	-0.607	9.158	-0.781	0.660	7.239	0.635	0.220
	(0.655)	(13.44)	(0.804)	(1.622)	(7.744)	(2.968)	(0.239)
D_URBAN	-0.0740	2.531	-0.273**	-0.147	-1.685	-1.200***	0.0442
	(0.0908)	(1.866)	(0.112)	(0.225)	(1.077)	(0.409)	(0.0331)
D_OPEN	-0.00295	0.0453	0.0012	0.0027	0.0083	0.0118	0.0002
	(0.00189)	(0.0386)	(0.0023)	(0.0047)	(0.0224)	(0.0085)	(0.0007)
TOT	-0.0017	0.0619	-0.0117**	-0.0004	-0.0171	-0.0020	0.0012
	(0.0042)	(0.0878)	(0.0052)	(0.0105)	(0.0507)	(0.0191)	(0.0015)
D_AGE14	-0.0664	1.474	-0.105	-0.499	-0.309	-0.510	-0.0626
	(0.133)	(2.705)	(0.165)	(0.330)	(1.568)	(0.600)	(0.0481)
AGE65	-0.0875	0.719	-0.198*	-0.0134	0.337	0.231	0.0007
	(0.0839)	(1.735)	(0.102)	(0.208)	(1.011)	(0.379)	(0.0303)
*D*_log(*POP*)	-1.617	-260.4**	13.36*	-9.048	-25.88	-89.44***	-9.011***
	(6.505)	(132.3)	(7.961)	(16.21)	(77.99)	(29.88)	(2.613)
D_UNEMPOY	-0.0117	0.269	0.0229	0.0122	0.370**	0.120*	-0.0047
	(0.0144)	(0.294)	(0.0176)	(0.0357)	(0.170)	(0.0649)	(0.0052)
TREND	0.0147	-0.283	0.0414*	0.0163	-0.0516	-0.0845	-0.0029
	(0.0182)	(0.377)	(0.0222)	(0.0450)	(0.222)	(0.0820)	(0.0066)
RECESSION	-0.0911	0.550	-0.0492	-0.0277	1.064	0.167	0.0116
	(0.0847)	(1.731)	(0.104)	(0.210)	(1.002)	(0.383)	(0.0309)
YEAR2003	0.0485	-1.240	0.595***	-0.109	1.298	0.990***	-0.0504*
	(0.0780)	(1.591)	(0.0955)	(0.194)	(0.922)	(0.355)	(0.0287)
Constant	0.511	-4.115	2.060***	-0.308	1.107	1.307	-0.0628
	(0.554)	(11.37)	(0.678)	(1.377)	(6.612)	(2.500)	(0.202)
Observations	32	32	32	32	32	32	32
R-squared	0.123	0.230	0.891	0.225	0.311	0.632	0.470

Note: Figures in parentheses are standard errors. *D_* denotes the first difference.
***, **, * indicate significance levels at 1%, 5%, and 10%.

Summary of the estimated coefficients for each political variable

(Dependent variable: Ministry budget share in total government budget)

Ministry\Political variable	*Non-Elected Government*	*MILITARY PM*	*Share of MP_i*	*$LEADER_i$*	*$OUTSIDER_i$*
Central Fund	0.319	-0.0109	4.046***		
(Dep. var: *Budget*)	(1.665)	(1.536)	(1.506)		
Office of the Prime Minister	-0.164	-0.222	-0.814***		
(Dep. var: *Budget*)	(0.268)	(0.245)	(0.253)		
Ministry of Defence	0.801**	0.635*	0.740**	0.187	-0.115
(Dep. var: *D_Budget*)	(0.342)	(0.325)	(0.330)	(0.137)	(0.175)
Ministry of Finance	0.715	-2.762***	1.544	-15.25	-0.0875
(Dep. var: *D_Budget*)	(1.151)	(0.958)	(1.232)	(11.28)	(0.331)
Ministry of Foreign Affairs	0.0734***	0.0433*	0.0899***	0.0439***	0.0188***
(Dep. var: *D_Budget*)	(0.0242)	(0.0226)	(0.0238)	(0.00802)	(0.00607)
Ministry of Agriculture and Cooperatives	-0.466**	0.152	-0.801***	-0.175***	-0.240**
(Dep. var: *D_Budget*)	(0.236)	(0.232)	(0.184)	(0.0645)	(0.104)
Ministry of Transport	-0.486	0.250	-0.900***	-0.549***	-0.132
(Dep. var: *D_Budget*)	(0.344)	(0.321)	(0.210)	(0.0542)	(0.127)
Ministry of Commerce	0.177***	-0.00176	0.180***	0.0824***	0.0701***
(Dep. var: *D_Budget*)	(0.0471)	(0.0521)	(0.0466)	(0.0252)	(0.0122)
Ministry of Interior	-1.647	-0.0135	-0.577	1.715***	-2.121***
(Dep. var: *D_Budget*)	(1.119)	(1.064)	(0.813)	(0.307)	(0.294)
Ministry of Justice	-0.0178	-0.150**	-0.0659	-0.0173	-0.0561
(Dep. var: *Budget*)	(0.0691)	(0.0588)	(0.0628)	(0.0267)	(0.0522)
Ministry of Science and Technology	-0.275**	0.0700	-0.176	-0.104**	-0.273***
(Dep. var: *D_Budget*)	(0.134)	(0.129)	(0.115)	(0.0420)	(0.0577)
Ministry of Education	0.264	1.034*	0.937*	0.296**	0.343*
(Dep. var: *D_Budget*)	(0.667)	(0.590)	(0.528)	(0.134)	(0.177)
Ministry of Public Health	-0.255	0.162	-0.355**	-0.0872*	-0.0755
(Dep. var: *D_Budget*)	(0.251)	(0.234)	(0.157)	(0.0457)	(0.136)
Ministry of Industry	-0.0120	0.0248	-0.0002	0.0138	-0.0753***
(Dep. var: *D_Budget*)	(0.0203)	(0.0183)	(0.0215)	(0.0143)	(0.0117)

Note: Figures in parentheses are standard errors. *D_* denotes the first difference.
***, **, * indicate significance levels at 1%, 5%, and 10%.

Robustness test

Ministry\Political variable	Non-Elected Government	MILITARY PM	Share of MP_i	$LEADER_i$	$OUTSIDER_i$
Central Fund	X	X	+		
Office of the Prime Minister	X	X	-		
Ministry of Defence	X	X	X	X	X
Ministry of Finance	X	-	X	X	X
Ministry of Foreign Affairs	X	+	+	+	+
Ministry of Agriculture and Cooperatives	-	X	X	-	-
Ministry of Transport	X	X	X	-	X
Ministry of Commerce	+	X	X	X	X
Ministry of Interior	X	X	X	+	X
Ministry of Justice	X	-	X	X	X
Ministry of Science and Technology	-	X	X	X	-
Ministry of Education	X	+	X	X	+
Ministry of Public Health	X	X	X	X	X
Ministry of Industry	X	X	X	X	-

Note:

X = The estimated coefficient does not remain robustly significant with the same sign.

+ = The estimated coefficient remains robustly significant with the same positive sign.

- = The estimated coefficient remains robustly significant with the same negative sign.

Key findings from the robustness test:

Non-Elected Government

- The relationships between a non-elected government and military-related expenditures are non-robust.

- The positive impact of a non-elected government on the budget share allocated to the Ministry of Commerce remains robustly significant.

- The negative impact of a '*non-elected*' government on the share of budgetary spending of the Ministry of Agriculture and Cooperatives and Ministry of Science and Technology are robustly significant. The robustness result hence underscores that science and technology development of Thailand in particular the agricultural development could be hindered by a post-coup government.

Key findings (continued):

MILITARY PM

- Prime ministers with high military ranking behave differently from their civilian counterparts. They robustly allocated more budget shares towards the Ministry of Foreign Affairs and Ministry of Education, while the Ministry of Finance and Ministry of Justice would receive less.

Share of MP

- As the share of MP of the leader party increases, government robustly allocates more budgets to the Central Fund for the prime minister and Ministry of Foreign Affairs, and less to the Office of the Prime Minister.

Key findings (continued):

LEADER

- The share of budgetary spending is positively influence as the administrative minister and the PM belongs to the same political party is robust for the Ministry of Foreign Affairs and the Ministry of Interior, but the negative influence also holds for the Ministry of Agriculture and Cooperatives and the Ministry of Transport respectively.

OUTSIDER

- For the Ministry of Agriculture and Cooperatives, Ministry of Science and Technology, and Ministry of Industry, non-parliamentarian ministers have a statistically significant negative impact on the share of public spending allotted to their ministries.

- For the Ministry of Foreign Affairs and Ministry of Education, their budget share increase significantly as the ministry is governed by an *outsider*.

Concluding remarks

- Military forces and coalition parties does have a significant influence on the expenditure budget allocation in Thailand. The impact could go beyond public consumptions of the present government into the science and technological development which is a key engine to drive the economy in the future.

- This study thus suggests that there should be some earmarks or safeguards for R&D expenditure so as to guard the science and technological development of the country from either military or coalition party influences in the future.

- For further study, the hypothesis on populist policy of a non-elected government should be investigated. The long term impact of government budget formulation on the sustainable development of the country is also worth a thorough examination.

THANK

YOU